

	# in the nth position
└─ l2BenfordProb.R	# Joint Benford probability of
	# digit pairs (Eq. 4.3)
└─ benfordDist.R	# Full Benford digit distribution
	# for a given position
└─ president_2016/	# 2016 U.S. Presidential Election
└─ elecData2016.rds	# Raw precinct-level data
	# (source: MIT MEDSL)
└─ dataRefiner.R	# Filters raw data to relevant
	# states/candidates/modes
└─ run2016.R	# Runs the full L2-BL 10 pipeline
	# by state
└─ resultsFormatter2016.r	# Compiles results, applies
	# Bonferroni correction, extracts
	# flagged counties
└─ president_2020/	# 2020 U.S. Presidential Election
└─ elecData2020.rds	# Raw precinct-level data
	# (source: MIT MEDSL)
└─ dataRefiner.R	# Filters raw data to relevant
	# states/candidates/modes
└─ run2020.R	# Runs the full L2-BL 10 pipeline
	# by state
└─ resultsFormatter2020.r	# Compiles results, applies
	# Bonferroni correction, extracts
	# flagged counties
└─ weibullSimulations/	# Weibull conformity to L2-BL 10
	# (Section 4.3)
└─ simWeibull.R	# Tests L2-BL 10 conformity across
	# a grid of Weibull params
└─ graphing/	# Figure generation (Figs. 1-7)
└─ fig1Generator.R	# Benford's Law first-digit
	# distribution
└─ fig2Generator.R	# Digit distributions for
	# positions 1-4
└─ fig3Generator.R	# Vote count distributions with
	# Weibull overlays
└─ fig4Generator.R	# d-KS test results across
	# parameter space
└─ fig5Generator.R	# Discrete Weibull PMFs across
	# parameterizations
└─ fig6Generator.R	# L2-BL 10 conformity across
	# Weibull parameter grid
└─ fig7Generator.R	# County precinct count
	# distributions

Usage

Running an election analysis

Each election directory (`president_2016/`, `president_2020/`) is run in sequence:

1. **Set working directory** to the election directory (e.g., `president_2016/`).
2. **Run `dataRefiner.R`** to filter raw data into a cleaned subset (`relData20XX.rds`). The nine states analyzed are FL, GA, MD, MO, NE, NY, PA, TN, and WA. Some states are restricted to in-person votes due to how absentee data was reported.
3. **Run `run20XX.R`** to execute the full L2-BL 10 pipeline. This produces per-state results files. This step is computationally intensive (1000 bootstrap d-KS samples + 100 Monte Carlo chi-squared simulations per county-candidate).
4. **Run `resultsFormatter20XX.r`** to compile state-level results, apply Bonferroni corrections, and extract fully flagged counties.

Running the Weibull simulations

Set the working directory to `weibullSimulations/` and run `simWeibull.R`. This tests L2-BL 10 conformity of the discrete Weibull across a grid of parameterizations (μ from 1 to 10, σ from 0.1 to 2) at sample sizes of 25, 50, 100, and 200.

Utility functions

The `utilities/` directory contains standalone functions that can be sourced individually. Note that `boot_dKS.R` and `estWeibullChiSq.R` depend on `dWeibullAlt` or the local `d/p/q/r` discrete Weibull functions.

Data

Precinct-level election data is from the [MIT Election Data and Science Lab](#):

- 2016: doi.org/10.7910/DVN/LYWX3D
- 2020: doi.org/10.7910/DVN/JXPRES